TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 17

5-MINUTE REVISION

Uniformity Trials

From the chapter's own revision points, formulae and questions.

Selected from Chapter 17 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Uniformity trial → used to study the nature and extent of fertility variation in a field.
- A single crop variety is grown over the entire field.
- All management inputs, including fertilizer if used, are applied uniformly.
- A substantial border is removed before dividing the field into basic units.
- Basic unit → smallest unit from which yield is separately recorded.

Quick revision · 2

- Yield differences among basic units → measure of soil heterogeneity.
- Smaller basic units → finer spatial detail, balanced against harvesting accuracy and cost.
- Uniformity trials can help determine → fertility variation, plot size, plot shape, and block size.
- Fertility contour map → graphical representation of soil heterogeneity.
- Moving averages → used to reduce random variation and identify areas of similar fertility.

Quick revision · 3

- Similar fertility areas → grouped together in the contour map.
- Serial correlation → measures the relationship between neighbouring observations.
- Horizontal and vertical serial correlations → used to identify the direction of fertility variation.
- Low serial correlation → little linear association at the chosen lag; this alone does not establish patches.
- High serial correlation → neighbouring yields are associated; use the field map to assess gradients and patches.

Quick revision · 4

- Mean square between strips → simpler method having the same objective as serial correlation.
- Between vertical strips → measures variation across columns, in the horizontal direction.
- Between horizontal strips → measures variation across rows, in the vertical direction.
- Larger strip mean square → indicates greater variation in that direction.
- Fairfield Smith's variance law → relates plot size to the variance of yield per basic unit area.

Quick revision · 5

- Smith's law → $V_x = V_1/x^b$.
- Smith's index of soil heterogeneity → b.
- b → quantitative measure of soil heterogeneity.
- Lower b → slower fitted variance reduction, often associated with positive nearby correlation; a gradient is not established.
- Higher b → a faster fitted reduction in variance as plot size increases; it does not establish a patch pattern.

Quick revision · 6

- Values from 0 to 1 describe the usual idealised range of Smith's index; empirical fitted values can fall outside this range.
- Simulated plots → formed by combining basic units into different sizes and shapes.
- Only plot arrangements that fit exactly within the whole field are used.
- Between-plot sum of squares → calculated for each simulated plot size and shape.
- Variance per basic unit area V_x → sample variance of plot means, equal to $Q_x/(rc - x)$.

Quick revision · 7

- Alternative shapes partition the same field → their variance estimates are dependent.
- Retain shapes separately or state a descriptive averaging rule; an ordinary F test does not justify pooling dependent partitions.
- Smith's index → estimated by regression of $Y = \log V_x - \log V_1$ on $X = \log x$.
- In $Y = cX$, $c = -b$.
- Adjusted b → obtained using interpolation from the adjusted b table.

Quick revision · 8

- Maximum curvature method → used to determine optimum plot size.
- In maximum curvature method → plot size is plotted on the X-axis and CV on the Y-axis.
- As plot size increases → CV generally decreases.
- Apparent bend → a traditional plot-size guide sensitive to axis scales and the available arrangements.
- In the rice example → the field was 12 m × 17 m.

Quick revision · 9

- After removing a 1 m border from all sides → effective area was 10 m × 15 m.
- With 1 m × 1 m basic units → there were 150 basic units.
- In the example, vertical serial correlation → 0.438627.
- In the example, horizontal serial correlation → 0.54317.
- Positive sequence correlations describe nearby similarity; they do not by themselves establish patches or gradients.

Quick revision · 10

- The higher horizontal sequence correlation alone does not establish a horizontal fertility gradient.
- In the example, the larger horizontal-strip mean square indicates greater yield variation across rows; a monotonic gradient is not established.
- For plot size 25 m² in the example → $V_{25} = 1750.14$.
- The calculated Smith's index in the example was $b = 0.3607$ before adjustment.
- The adjusted Smith's index in the example was approximately 0.41156.

Quick revision · 11

- The index summarises variance reduction with plot size; identifying gradients or patches also requires spatial information.
- Uniformity trials are important because field variability exists even when all management practices are kept uniform.

Important formulae · 1

Serial correlation:

$$r_s = \frac{\sum_{i=1}^{n} X_i X_{i+1} - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}}{\sum_{i=1}^{n} X_i^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}}$$

Total number of basic units:

$$n = r \times c$$

Vertical sum of squares:

$$SS_V = \frac{\sum_{i=1}^{c} V_i^2}{r} - \frac{G^2}{n}$$

Important formulae · 2

Horizontal sum of squares:

$$SS_H = \frac{\sum_{i=1}^{r} H_i^2}{c} - \frac{G^2}{n}$$

Vertical mean square:

$$MS_V = \frac{SS_V}{c-1}$$

Horizontal mean square:

$$MS_H = \frac{SS_H}{r-1}$$

Important formulae · 3

Fairfield Smith's variance law:

$$V_x = \frac{V_1}{x^b}$$

Logarithmic form of Smith's variance law:

$$\log V_x = \log V_1 - b \log x$$

Difference form:

$$\log V_x - \log V_1 = -b \log x$$

Important formulae · 4

Regression form:

$$Y = cX$$

where

$$Y = \log V_x - \log V_1$$

and

$$X = \log x$$

Relationship between regression coefficient and Smith's index:

$$c = -b$$

Important formulae · 5

Number of simulated plots:

$$w = \frac{rc}{x}$$

Between-plot sum of squares:

$$Q_x = \sum_{i=1}^{w} \frac{T_i^2}{x} - \frac{G^2}{rc}$$

Variance per unit area:

$$V_x = \frac{Q_x}{rc - x}$$

Important formulae · 6

Coefficient of variation:

$$CV = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

Standard deviation from variance:

$$SD = \sqrt{V_x}$$

Adjusted Smith's index:

$$\text{Adjusted } b = y_1 + (b_{\text{cal}} - L_1)\frac{y_2 - y_1}{L_2 - L_1}$$

Important formulae · 7

Descriptive ratio for two plot shapes (not an F test for dependent partitions):

$$R = \frac{\text{Larger between-plot sum of squares}}{\text{Smaller between-plot sum of squares}}$$

Self-check from the chapter

1. A field trial conducted under uniform treatment to study field variability is called a _____ trial. **[Uniformity]**
2. The smallest harvested subdivision in a uniformity trial is called a _____ unit. **[Basic]**
3. A map showing spatial patterns in yield is called a fertility _____ map. **[Contour]**
4. The correlation between yields of neighbouring units is a form of _____ correlation. **[Serial]**

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